FAST ALGORITHMS FOR THE LABEL PROPAGATION OPERATOR ON SIGNED GRAPHS

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FUDAN UNIVERSITY

NEURAL INFORMATION PROCESSING SYSTEMS

Signed multi-rooted tree sampling with Rao-Blackwellized variance reduction

Problem and motivation

Signed graphs model cooperative and antagonistic interactions. Gaussian Random Fields infer unknown labels from a small fixed labeled set. Leader-follower DeGroot dynamics predicts follower opinions from stubborn leaders. Both solve the same discrete Dirichlet problem on signed graphs, where the label propagation operator Q maps labeled or leader states to unlabeled or follower states.

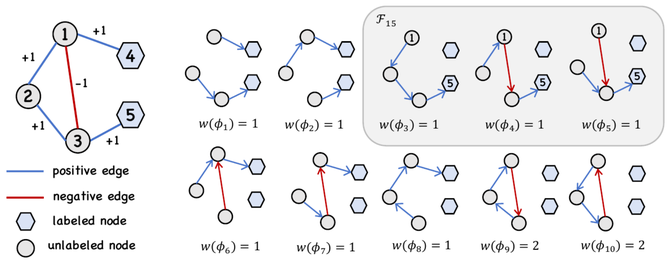
$$\boldsymbol{f}_U = -L_{U,U}^{-1} L_{U,L} \boldsymbol{y}_L, \qquad Q = -L_{U,U}^{-1} L_{U,L}$$

Direct inversion has cubic cost in the number of unlabeled nodes. Random-walk estimators avoid inversion but have high variance and may become biased after truncation.

Contributions

- Signed Multi-Rooted Spanning Trees (SMRSTs) extend tree interpretations to graphs with negative edges.
- MultiTreeQ samples SMRSTs with importance weights that correct the contribution of negative cycles.
- Rao-Blackwellization averages all possible boundary edges and provably lowers estimator variance.

Signed multi-rooted spanning trees



(a) Signed Graph (b) All valid Signed Multi-Rooted Spanning Tree (SMRST) set $\mathcal{F}$

Each unlabeled node has one outgoing edge, each component has one labeled root, and negative cycles remain valid. The SMRST weight is

$$W(\phi) = 2^{c(\phi)} \prod_{e \in \phi} |W_e|$$

Tree interpretation of Q

For unlabeled node i and labeled root j, each operator entry equals an expectation under the target SMRST distribution:

$$Q_{ij} = \mathbb{E}_{\phi \sim P}[\mathbf{1}\{r_\phi(i) = j\} \operatorname{sgn}(P_{i \to j})]$$

Standard loop-erased walks sample only according to absolute edge weights and therefore under-sample negative cycles. Self-normalized importance sampling restores the missing cycle-dependent factor.

Why Rao-Blackwellization helps

A sampled tree component can reach many labeled nodes through its boundary. MultiTreeQ replaces one sampled absorption edge with the conditional expectation over every boundary edge. This removes variance without changing the target expectation.

$$\mathbb{E}[\widehat{Q}_{ij,\mathrm{RB}}] = \mathbb{E}[\widehat{Q}_{ij}], \qquad \mathrm{Var}(\widehat{Q}_{ij,\mathrm{RB}}) \le \mathrm{Var}(\widehat{Q}_{ij})$$

MultiTreeQ algorithm

Precompute boundary capacities, sample signed loop-erased forests, keep negative cycles, erase positive cycles, and aggregate conditional boundary expectations.

1 Boundary preprocessing

For every unlabeled node, compute its total absolute connection to the labeled set. These values support constant-time access to the boundary mass during sampling and avoid repeated scans of labeled neighbors.

2 Signed Wilson sampling

Start with labeled nodes as absorbing roots. From each unabsorbed node, run a random walk using absolute edge weights. A positive cycle is erased, following Wilson's algorithm. A negative cycle is retained, its nodes become absorbed, and the sample's cycle count increases.

3 Importance weighting

Ordinary loop-erased walks sample a forest in proportion to the product of absolute edge weights. MultiTreeQ adds the cycle-dependent importance weight from the SMRST distribution, correcting the under-sampling of negative cycles.

4 Component-wise aggregation

For each valid tree component, choose a reference node and propagate path signs through the branch. Aggregate every edge from the component to every adjacent label, then update all operator entries for nodes in that component. One sampled tree therefore updates many Q entries at once.

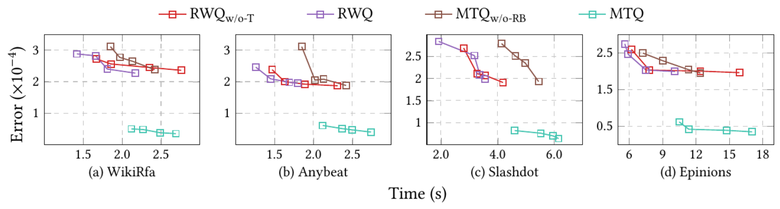
Rao-Blackwellized estimator

After removing boundary edges, MultiTreeQ takes the component containing the queried node and marginalizes over every edge from that component to labeled nodes:

$$\widehat{Q}^{(k)}_{ij,\mathrm{RB}} = \frac{\sum_{u \in T_i} \operatorname{sgn}(P_{i \to u}) W_{uj}}{\sum_{u \in T_i} \sum_{j' \in V_L} |W_{uj'}|} \, 2^{c(\phi_k)}$$

The estimator remains unbiased and inherits the finite-sample error guarantee of the original self-normalized estimator.

Accuracy-time trade-off



MTQ forms a distinct low-error frontier on WikiRfa, Anybeat, Slashdot, and Epinions. The added runtime remains modest relative to the reduction in MAE.

Complexity

Boundary preprocessing is linear in the number of edges. The sample count gives an entrywise error guarantee.

$$\mathbb{E}[T_{\mathrm{MultiTreeQ}}] = O(|E| + T)$$

$$\ell = \left\lceil 2 \left(\frac{\alpha}{\beta}\right)^2 \left(\frac{\epsilon+2}{\epsilon}\right)^2 \ln \frac{4}{\delta} \right\rceil, \qquad \Pr\left(|\widehat{Q}_{ij} - Q_{ij}| \le \epsilon\right) \ge 1 - \delta$$

Experimental setup

Seven small and medium signed networks evaluate accuracy. Five large networks evaluate runtime. Exact inversion provides small-graph ground truth, while conjugate gradient covers medium graphs. Every MAE result reports five runs.

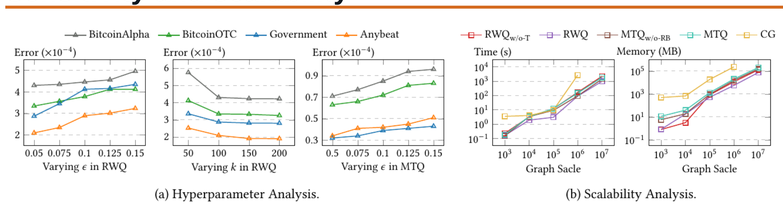
Accuracy on real networks

Table 1: Mean Absolute error ($\times 10^{-4}$) of our algorithms on six small datasets.

Dataset	Node	Edge	$\mathrm{RWQ}_{\mathrm{w/o\text{-}T}}$	RWQ	$\mathrm{MTQ}_{\mathrm{w/o\text{-}RB}}$	MTQ
BitcoinAlpha	3,783	24,186	3.90±0.03	4.30±0.02	4.25±0.08	**0.96±0.01**
BitcoinOTC	5,881	35,592	3.24±0.04	3.34±0.04	3.38±0.02	**0.83±0.01**
Government*	7,057	89,455	2.76±0.03	2.87±0.03	2.89±0.01	**0.43±0.00**
WikiRfa	11,017	170,335	2.56±0.01	2.82±0.01	2.77±0.03	**0.48±0.00**
Anybeat*	12,645	49,132	2.00±0.02	2.09±0.02	2.05±0.01	**0.51±0.01**
Slashdot	82,140	549,202	2.11±0.01	2.52±0.02	2.51±0.02	**0.76±0.01**
Epinions	131,828	841,372	2.03±0.01	2.46±0.01	2.28±0.02	**0.42±0.01**

MTQ has the lowest MAE on every dataset. Relative to RWQ, the error falls by 2.8× to 6.4× and remains below 10^{-4} in all seven cases.

Sensitivity and scalability



(a) Hyperparameter Analysis. (b) Scalability Analysis.

Smaller ε improves accuracy by increasing sampling rounds. The truncation effect saturates near k = 150. MTQ time and memory scale linearly from 10^3 to 10^7 synthetic nodes.

Runtime on large networks

Dataset	Node	Edge	$\mathrm{RWQ}_{\mathrm{w/o\text{-}T}}$	RWQ	$\mathrm{MTQ}_{\mathrm{w/o\text{-}RB}}$	MTQ
TwitterSG	753,944	12,848,093	114.3	63.29	61.13	92.06
Youtube*	1,134,890	2,987,624	186.8	96.79	93.78	147.3
USPatents*	3,774,768	16,518,947	817.2	497.3	270.2	381.5
Dblp*	7,577,304	12,282,059	1678	892.5	823.6	1026
Livejournal*	10,690,276	112,307,385	3078	2145	2031	2540

LiveJournal*: 10,690,276 nodes and 112,307,385 edges. MTQ finishes in 2,540 s. On the synthetic 10^6-node graph, CG needs about 1,341 s and 235 GB, then fails at larger scales.

Main findings

- Variance reduction drives the accuracy gain. MTQ without Rao-Blackwellization performs similarly to random-walk baselines.
- Sampling an SMRST reuses overlapping random-walk trajectories and avoids repeated local exploration.
- The current aggregation cost grows with the labeled set. Hierarchical grouping and dynamic signed graphs are natural next steps.

Takeaway: tree sampling plus boundary averaging makes explicit signed label propagation accurate and scalable.